

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: Group 30

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. Project description

2.a) Domain of the Application:

The domain of our application is academic communication and research discussion. The platform is designed to help researchers and scholars share insights, discuss trending topics, and seek help from peers in their respective academic fields. Researchers often read numerous scientific papers, leading to new questions, ideas, and discussions. However, finding others with the same background to discuss these topics can be challenging, which slows down research progress.

Our application aims to bridge this gap by providing a centralized platform for academic discussions. Users can post and discuss their research insights, exchange opinions, and engage in collaborative discussions. Additionally, the platform will be divided into different academic fields (e.g., Computer Science, Physics, Biology), allowing users to find and participate in relevant discussions efficiently. We also support interest-based groups, enabling researchers with similar research interests to connect more effectively.

2.b) Modeled Aspects of the Domain:

The database will manage various aspects of the platform, including:

- User Information: Each user will have a unique User ID, email, and other profile details.
- Reading List: Users can bookmark posts they intend to read later.
- Academic Fields: Posts will be categorized into different academic fields, ensuring structured discussions.
- Posts & Discussions:
 - Each post will have a post ID, content, number of likes, timestamp, and the academic field it belongs to.

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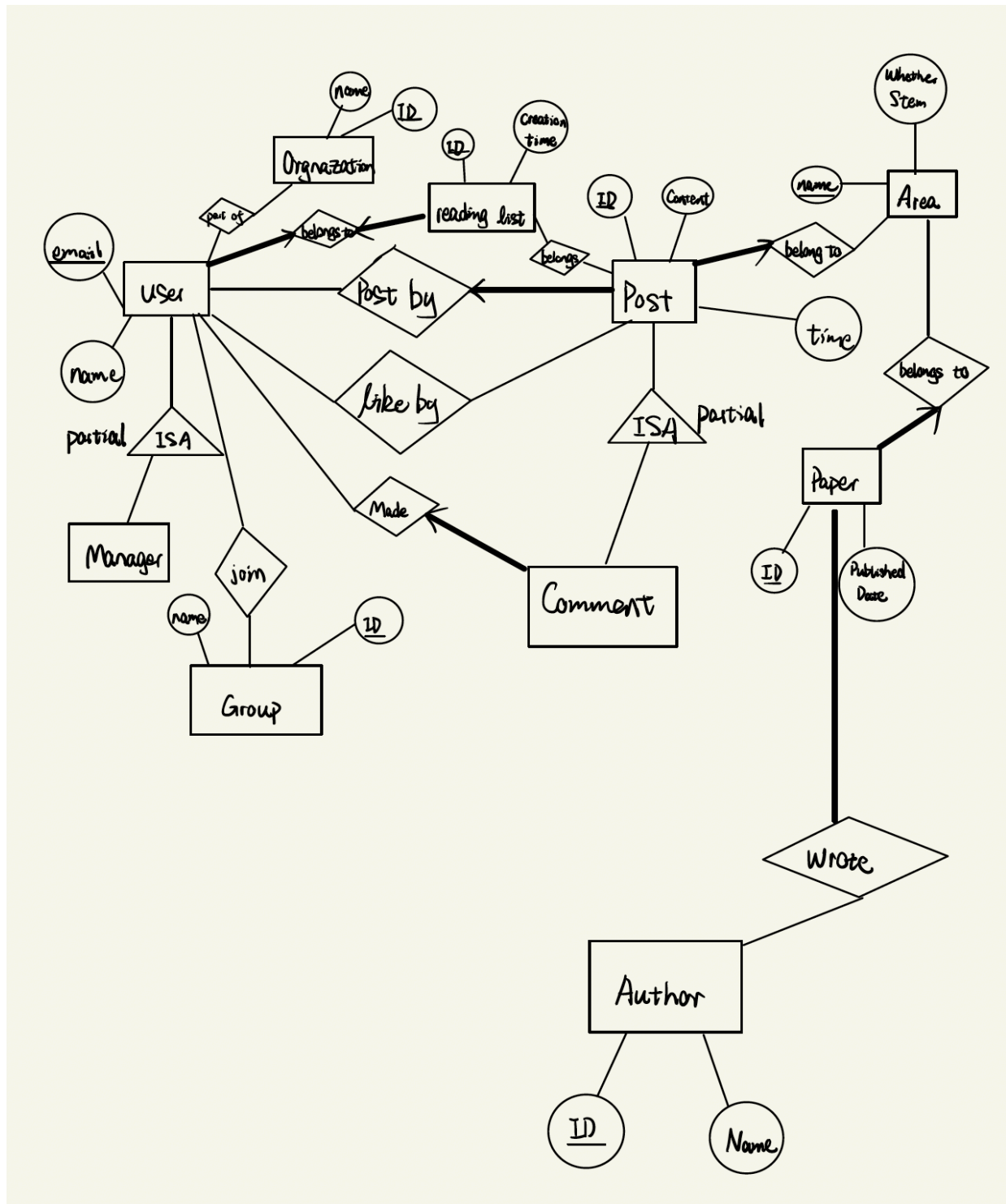
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- Posts will be associated with users who created them.
- Each post will contain comments, which will include the comment content, creator ID, and timestamp.
- Interest Groups: Users can join groups based on research interests. The system will track group members and group discussions.
- Administrative Roles: Certain users will have moderator privileges to manage group discussions and platform activities.

3. Database Specifications:

The database will store user accounts, posts, comments, groups, and interactions for academic discussions. Users will be able to post research insights, comment on discussions, and join interest groups. Each post will be categorized under a specific academic field, allowing for structured and relevant discussions. Additionally, the database will track reading lists and moderation roles. An author can also write an essay, which will be stored and associated with their user account. The system will be designed to efficiently query and manage large-scale discussions, ensuring that academic discourse remains accessible and well-organized.

4. ER Diagram for the database (hand drawn):



5. Explanation for the ER diagram

- Number of entities:

From the diagram, it includes 8 entities: User, Post, Sports, Group, Organization, Reading List, Paper and Author.

- Primary Keys:

Every entity has their own primary keys: User(email), Post(ID), Area(name), Group(ID), Organization(ID), Reading List(ID), Paper(ID), Author(ID).

- Attributes:

Each entity has at least one non-primary key attribute: User(name), Post(content, time), Area(Whether stem), Group(name), Organization(name), Reading List(creation time), Paper(Published Date) and Author(name).

- Suitable relationships:

post **is posted by** an User; a post can **be liked** by a user; a post **is belong to** area block; a user can **join** a discussion group; a user **can be a part of** an organization; a reading list **is belong to** an user; a post **is belong to** a reading list; a paper **is belong to** an area; an author **wrote** a paper.

- ISA:

First ISA:

Superclass (Parent Entity): User

Subclass (Child Entity): Manager

ISA Relationship: Manager ISA User(partial participation)

Second ISA:

Superclass (Parent Entity): Post

Subclass(Child Entity): Content

ISA Relationship: Content ISA User(partial participation)

- Weak Entity: **The weak entity is not suitable for our system. Instead, we added two ISA relationships, which are shown in the ER diagram and explained above.**

6. Acknowledgement about Using AI tools

Yes, we confirm that we **only use one** AI tool, ChatGPT, for grammar revision in our deliverables.